Lesson 1.4: Context limits and memory

Claude Certified Associate – Foundations (CCAO-F) · Module 1: Platform & Model Foundations · Lesson 4 (the last lesson of Module 1) · 13:32

Description

A chat can go wrong in two opposite ways. Keep one chat going forever, and it gets crowded: early details slip. Start fresh every time, and you explain everything again. This lesson explains what Claude's context window is, how big it is, why a long chat loses focus, and what Claude does when a chat fills up. Then it gives you four moves, keep going, restart, summarize then restart, or persist, and four questions that pick the right one. You see where Anthropic publishes these facts: the official exam guide, the Claude Help Center, the Claude Platform Docs and Claude Academy. Then a real working chat for this course is handed to a fresh chat with a short hand-off summary, and every carried decision is checked. Along the way you see what a restart alone gives you (an incognito chat that knows nothing), and how to choose what Claude remembers. The lesson ends with one exam-style scenario, reasoned out loud, and the cue to the Module 1 quiz.

Learning outcomes

By the end of this lesson you can:

1. Explain what a context window is, and what happens when a chat gets long.
2. Decide when to keep going, restart, summarize, or persist.
3. Hand a working chat to a fresh one, and choose what Claude remembers.

Chapters

Time	Chapter
00:00	Preview and outcomes
00:37	Why it matters
01:25	The desk and four moves
03:16	Straight from the source
04:58	Hands-on: hand off a working chat
11:42	Exam angle
12:40	Recap and Module 1 quiz

The desk: a picture of the context window

The desk is this course's picture, not Anthropic's. Anthropic calls the context window the model's "working memory".

- **Everything on the desk is what Claude can see** in one chat: your messages, your files, its own replies, and the results of any tools it uses. What isn't on the desk, Claude can't see.
- **It's a big desk.** On paid plans, chats with Sonnet 5.5 and Opus 5.5 have a context window of 1 million tokens. Outside these models (Haiku 4.5, for example) it's 200,000 tokens, "about 500 pages of text or more". A token is a small chunk of text.
- **Bigger isn't tidier.** As a chat grows, accuracy and recall degrade. Anthropic's developer docs call this *context rot*: "more context isn't automatically better". Think of a rule from your first message, buried under fifty pages.
- **When the desk nearly fills,** Claude makes room by itself (on paid plans, with code execution on): it summarizes earlier messages and keeps going. Your full history is kept. The catch: longer chats that trigger this use more of your usage limit.
- **A new chat is an empty desk.** In Claude Academy's words, "The model doesn't learn from your corrections. It only responds to what's currently in context." Whatever the next chat needs, you carry over.

Four ways to carry context across:

Carrier	What it carries	Where
Hand-off summary (course term)	A short note you write or ask for, then paste into the new chat	Any chat
Project	Instructions and knowledge, within reach in every chat inside the Project; each Project has its own memory space	All plans (Free: up to five Projects)
Memory	Topics Claude saves about you as you chat, carried into new chats. You review them in Settings > Memory, under Topics, where you can edit or delete them	All plans; on by default for Free, Pro and Max
Chat search	Ask Claude, and it searches your past chats	Paid plans

Leaving no trace:

	Saved to your chat history?	Uses memory?	Saved to memory?
Normal chat	Yes	Yes (if memory is on)	Yes (if memory is on)
Memory off for one chat (+ menu → Memory, before the first message)	Yes, with a crossed-out memory icon by its title	No	No
Incognito chat (ghost icon, top right of a new chat)	No	No	No

The four questions (restart, summarize, or persist)

This decision flow is the course's own, built from Anthropic's guidance. It is not an Anthropic document.

1. **Same job, still on track?** Keep going.
2. **If not, does the next step need what this chat decided?** No: restart. Yes: summarize, then restart.
3. **Is it needed in every future chat?** Persist it: a Project for one body of work, Memory for things about you.
4. **Should it leave no trace?** Use an incognito chat, which isn't saved at all.

Signals that it's time to move: a phase of the work ends; Claude misses details from earlier in the chat; the chat is long and your usage matters; or, rarely, a length-limit message. On a desk this big, the signal is rarely the limit. It's usually a finished phase.

Straight from the source (what the lesson shows)

- **Exam guide, page 5:** Domain 3 (12% of scored items) ends with the bullet "Understand and manage context limitations and memory considerations (when to restart, summarize, or persist)". This course labels it task 3.4. The guide lists its task statements as unnumbered bullets, so 3.1 to 3.4 are this course's labels, not the guide's.
- **Exam guide, page 3:** the recommended experience includes "A practical understanding of AI limitations, including hallucinations, context constraints, and data sensitivity".
- **Help Center, context window sizes:** the Chatting table (Sonnet 5.5: 1M tokens) and the line below it (200K, about 500 pages, outside these models).
- **Help Center, automatic context management:** "When your conversation approaches the context window limit, Claude summarizes earlier messages to make room for new content." Longer conversations that trigger it use more of your usage limit. Your full history is preserved, and "You may occasionally notice Claude "organizing its thoughts"". Code execution must be on.
- **Claude Platform Docs, context windows:** the window is the model's "working memory"; "more context isn't automatically better"; accuracy and recall degrade (context rot).
- **Claude Academy, Working Memory:** the corrections line above, and "Knowing how the window works tells you how to structure context, when to front-load, and when to start fresh."
- **Help Center, memory:** the account switches in Settings > Memory, "Generate memory from chats" and "Search and reference chats", shown through the Help Center's own screenshots; Topics; turning memory off for a single chat.
- **Help Center, incognito chats:** "temporary conversations that aren't saved to your chat history or to Claude's memory".

Key terms

- **Context window:** everything Claude can see in one chat. Anthropic calls it the model's working memory.
- **Context rot:** more context, less focus. As a chat grows, accuracy and recall degrade.
- **Automatic context management:** near the limit, Claude summarizes earlier messages so the chat can continue (paid plans, code execution on). It uses more of your usage limit.
- **Hand-off summary:** a short note that carries one chat's decisions into the next (this course's term).
- **Memory:** what Claude saves about you, across chats, as individual topics you can review in Settings > Memory.
- **Incognito chat:** a chat that isn't saved to your history or to memory, and doesn't use memory.

What happened in the hands-on

Claude Desktop 2.9939.4 on Linux, Chat view, Sonnet 5.5 at Medium, recorded on 2026-10-04 on a real account. **The real job:** planning this course's Module 1 quiz. Phase one ran in a working chat; phase two ran in a fresh chat that got nothing but the hand-off.

1. **The working chat.** Before the first message, Memory was switched off for that chat only (+ menu), and Web search was unticked. The connected apps had been switched off before recording. Four files went on the desk: the official exam guide and the lesson notes of Lessons 1.1, 1.2 and 1.3, about ten thousand words in all, a small fraction of the window.
2. **Prompt 1** asked for the ideas a quiz question could fairly test in each lesson, each tagged with its task statement (3.1 to 3.4), and a flag on anything the lessons teach that the guide doesn't test.
3. **Claude's reply.** Its first line said it had the guide and the Lesson 1.1 notes, but the other two notes "didn't come through in context", so it read them from disk (it ran commands to read the files). Then it raised a true catch: the guide doesn't number its task statements, and the Lesson 1.1 notes made the numbers sound official. (That line has since been reworded in the Lesson 1.1 notes.) It gave one table per lesson. Every tag was checked against page 5 of the guide and held: features under 3.1, tiers under 3.2, matching a model under 3.3. Its flags, "Not in the guide at all": entry point and the four-decisions framework (the prep course's terms); Cowork and hand-offs (Cowork handing off a job, not this lesson's hand-off summary); effort levels, prices and timings, plan details, dated product news. Then skills from other domains, and a gap: task 3.4 lives in this lesson.

4. **Prompt 2, the decision and the hand-off.** Anything flagged stays out; 12 original questions, mixing multiple choice and multiple response; three of them go to Lesson 1.4. That decision existed only in this chat. In the same message: "This chat has done its job", then a request for a hand-off summary under 250 words, with the standing rules marked.
5. **The hand-off** (in the student kit) has the goal, the decisions, the testable ideas with tags, a "Standing rules (every quiz chat)" section and the next step. **One honest check:** the copied text counts 252 words against the "under 250" asked. Close, but over. Check the output against the ask.
6. **A restart alone.** A new incognito chat was asked how many questions the Module 1 quiz has and what was left out. It answered that it didn't know: it was the start of the conversation, with no memory of past chats, and it would rather say so than guess. That's the right answer, and it shows a new desk is empty.
7. **The fresh chat.** A new, normal chat. Memory switches back on in every new chat, so it was switched off again (the Web search tick carries over between chats). The hand-off was pasted with Ctrl+V and arrived as clean, formatted text, followed by one task line: draft the blueprint for the 12 questions as a table, no questions yet.
8. **The blueprint** (in the student kit), checked against the hand-off: 12 rows, yes. Three for Lesson 1.4 under 3.4, marked provisional, yes. Eight multiple choice and four multiple response, each saying how many to pick, yes. Nothing flagged (no Cowork, no entry point, no effort, no prices), yes. It even listed open points, like rows 10 to 12 waiting for the Lesson 1.4 notes. The crossed-out memory icon by the chat's title confirmed memory was off.
9. **Persist.** The hand-off's standing rules apply to every future quiz chat, so pasting them each time is the same setup, over and over. They belong in a Project's instructions. Lesson 5.1 builds that Project.

The exam link

Domain 3, Product and Model Selection (12% of scored items). This lesson serves the bullet this course labels **3.4**: "Understand and manage context limitations and memory considerations (when to restart, summarize, or persist)". The guide lists its task statements as unnumbered bullets; the numbers 3.1 to 3.4 are this course's labels. Lesson 1.1 introduced the four decisions before the first prompt (entry point, features, model, context), Lesson 1.2 covered features, Lesson 1.3 covered models, and this lesson covers context, the last of the four.

The lesson's practice scenario (original, written for this course, not from the exam). A marketing coordinator has planned a product launch in one chat for three weeks. Now Claude mixes up the old and new launch dates, and misses the tone rules from week one. She runs a launch every quarter. What's the most appropriate action? A: keep going, and repeat the tone rules in every message. B: get a summary of the current decisions, start a new chat with it, and put the tone rules in a Project's instructions. C: switch to a model with a bigger context window. D: move to an incognito chat, so old details can't interfere. The answer is **B**. Run the four questions: still on track? No. Does the next step need the decisions? Yes: summarize, then restart. Needed every time? The tone rules are: persist them in a Project. A keeps the cluttered desk. C buys a bigger desk, but the problem is clutter, not size. D throws the decisions away.

Module 1 quiz

That's Module 1 complete: the four decisions before the first prompt, from Lesson 1.1 to Lesson 1.4. Next up: the Module 1 quiz, which covers all four lessons (Domain 3, the task statements this course labels 3.1 to 3.4). There's no separate quick check for this lesson; the module quiz takes its place. The exam-style question in this lesson, the quick checks, the module quizzes and the practice exams are all original, written for this course against the official blueprint. They are never copies of the real exam.

Sources (all checked 2026-10-04)

- **Anthropic, *Claude Certified Associate – Foundations Exam Guide*.** Version 1.0, effective July 2026. §4 recommended experience (p. 3), §6 domain weights (p. 4) and Domain 3 task statements (p. 5), §7 How to Prepare (p. 6). Official download: https://everpath-course-content.s3-accelerate.amazonaws.com/instructor/6nizmqk8tpzpfjvt6qmmav7rh/public/1783542847/Claude+Certified+Associate+%E2%80%93+Foun
- **Claude Help Center:**
  - *How large is the context window on paid Claude plans?* (sizes; automatic context management; code execution): https://support.claude.com/en/articles/8606394-how-large-is-the-context-window-on-paid-claude-plans
  - *How do usage and length limits work?* (length limits; starting a new conversation): https://support.claude.com/en/articles/11647753-how-do-usage-and-length-limits-work
  - *Usage limit best practices* (conversation length affects usage): https://support.claude.com/en/articles/9797557-usage-limit-best-practices
  - *Use Claude's chat search and memory to build on previous context* (memory topics, Settings > Memory, project memory, memory off for a single chat, chat search): https://support.claude.com/en/articles/11817273-use-claude-s-chat-search-and-memory-to-build-on-previous-context
  - *Use incognito chats*: https://support.claude.com/en/articles/12260368-use-incognito-chats
  - *What are projects?*: https://support.claude.com/en/articles/9517075-what-are-projects
  - *Release notes* (context window compaction, November 24, 2025; memory updates, July 10 and August 25, 2026): https://support.claude.com/en/articles/12138966-release-notes
- **Claude Platform Docs,** *Context windows* (working memory, context rot): https://platform.claude.com/docs/en/build-with-claude/context-windows
- **Claude Academy,** *AI capabilities and limitations*, *Working Memory*: https://academy.claude.com/courses/ai-capabilities-and-limitations/working-memory
- **Anthropic's official CCAO-F prep course,** Course 1, *Claude Platform & Model Foundations* (learning objective: manage context limitations and use memory features to maintain continuity across sessions).

- **The app itself:** Claude Desktop 2.9939.4 on Linux, Chat view, the + menu (Memory, Web search) and incognito chats, observed 2026-10-04. Menus and labels can change between versions.

The desk picture, the four questions and the term "hand-off summary" are this course's own teaching tools, built from the sources above.

What you see in the lesson is a real account. Settings → Account and Settings → Memory were never opened; the Settings > Memory switches are shown through the Help Center's own screenshots. For the recording, Memory was switched off for each chat, Web search was unticked and the connected apps were switched off; all were restored afterwards.